DHS4_Script_Bender_Carliez_Guichard_Rabache

September 2, 2026

[ ]:
```
# Step 1: Install necessary library for handling Excel files.
# !pip is used to install library

!pip install pandas matplotlib

# Step 2: Import necessary libraries
import pandas as pd
from google.colab import files
import matplotlib.pyplot as plt

# Step 3: Upload the Financial Data File (Excel or CSV)
print("Please upload DHS4_Data_Bender_Carliez_Guichard_Rabache.csv")
uploaded = files.upload()  # Allows users to upload a file from their local␣
↪system

# Get the uploaded file name
file_name = list(uploaded.keys())[0]   # 0: picks the first filename
print(f"\nUploaded file: {file_name}")

# In your Topic 1 -> Dataset: you have "data.csv", "data.xlsx", "sample_prices.
↪csv"
# Import any one file or import multiple files in different run.

# Determine the file type and read the file
if file_name.endswith('.csv'):
    df = pd.read_csv(file_name)  # Read CSV file
elif file_name.endswith('.xlsx'):
    df = pd.read_excel(file_name)  # Read Excel file
else:
    raise ValueError(
        "Unsupported file format. "
        "Please upload a CSV or Excel file."
    )
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (3.10.0)
```

```
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-
packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Please upload DHS4_Data_Bender_Carliez_Guichard_Rabache.csv

<IPython.core.display.HTML object>

Saving DHS4_Data_Bender_Carliez_Guichard_Rabache.csv to
DHS4_Data_Bender_Carliez_Guichard_Rabache (1).csv

Uploaded file: DHS4_Data_Bender_Carliez_Guichard_Rabache (1).csv
```

[ ]:
```
# 1. CONVERSION IN DATAFRAME + cleaning
# (Session 2: df.dropna, subsetting)

# df = pd.DataFrame(data_list, columns=["Ticker", "Debt_to_Equity", "Return",␣
↪"Risk", "Revenue"])
df = df.dropna()   # remove lines with missing values


# 2. REMOVE OUTLIERS WITH IQR
# (Session 3: aggregation + vectorisation with pandas)

Q1 = df["Debt_to_Equity"].quantile(0.25)
Q3 = df["Debt_to_Equity"].quantile(0.75)
IQR = Q3 - Q1
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lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df_clean = df[(df["Debt_to_Equity"] >= lower) & (df["Debt_to_Equity"] <= upper)]


# 3. VISUALISATION (Session 4: Tell the story with data)


# BOX PLOT (with outliers)
plt.figure(figsize=(5,5))
plt.boxplot(df["Debt_to_Equity"])
plt.title("Boxplot du Debt/Equity (with outliers)")
plt.ylabel("Debt to Equity Ratio")
plt.show()

#Histogram

df_clean['Debt_to_Equity'].plot(kind='hist', bins=20, title='Debt_to_Equity')
plt.gca().spines[['top', 'right',]].set_visible(False)

#  SCATTER PLOTS (correlation with D/E)
for col in ["Return", "Risk", "Revenue"]:
    correlation = df_clean["Debt_to_Equity"].corr(df_clean[col])
    plt.figure(figsize=(5,5))
    plt.scatter(df_clean["Debt_to_Equity"], df_clean[col])
    plt.title(f"Debt/Equity vs {col}")
    plt.xlabel("Debt to Equity")
    plt.ylabel(col)
    plt.text(
        0.05,
        0.95,
        f'Correlation: {correlation:.2f}',
        horizontalalignment='left',
        verticalalignment='top',
        transform=plt.gca().transAxes,
        fontsize=10,
        bbox=dict(boxstyle='round,pad=0.5', fc='wheat', alpha=0.5)
    )
    plt.show()
#AI to write the value of the correlation directly on the scatter plot (the last 9 lines )


# 4. Descriptive statisticts
summary_stat=df.describe()
print(summary_stat)
```

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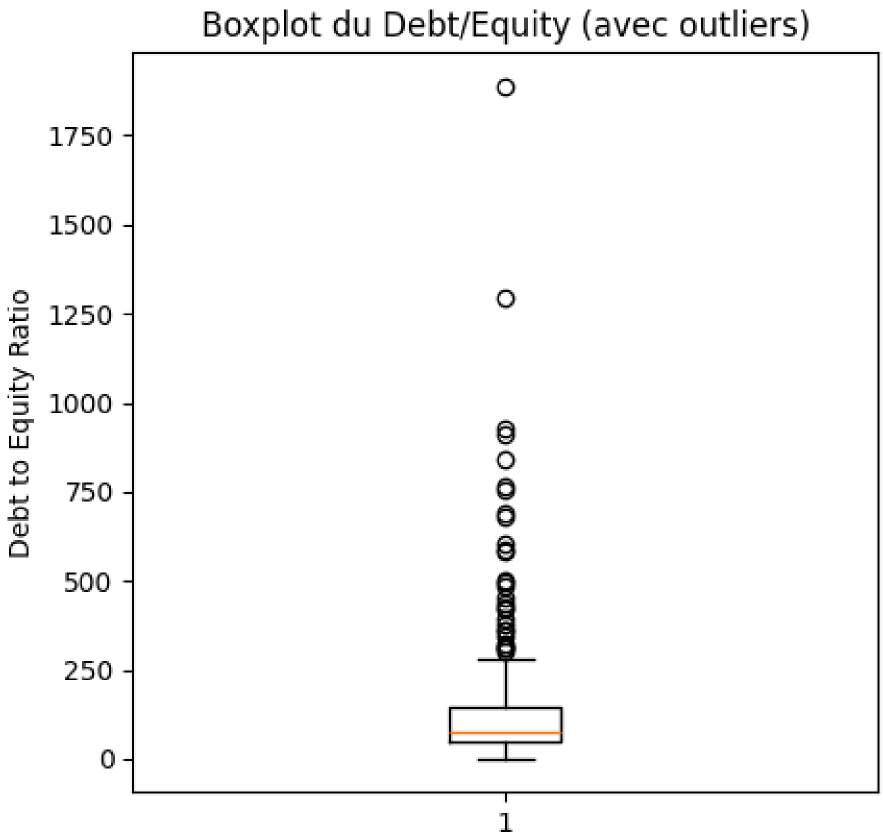
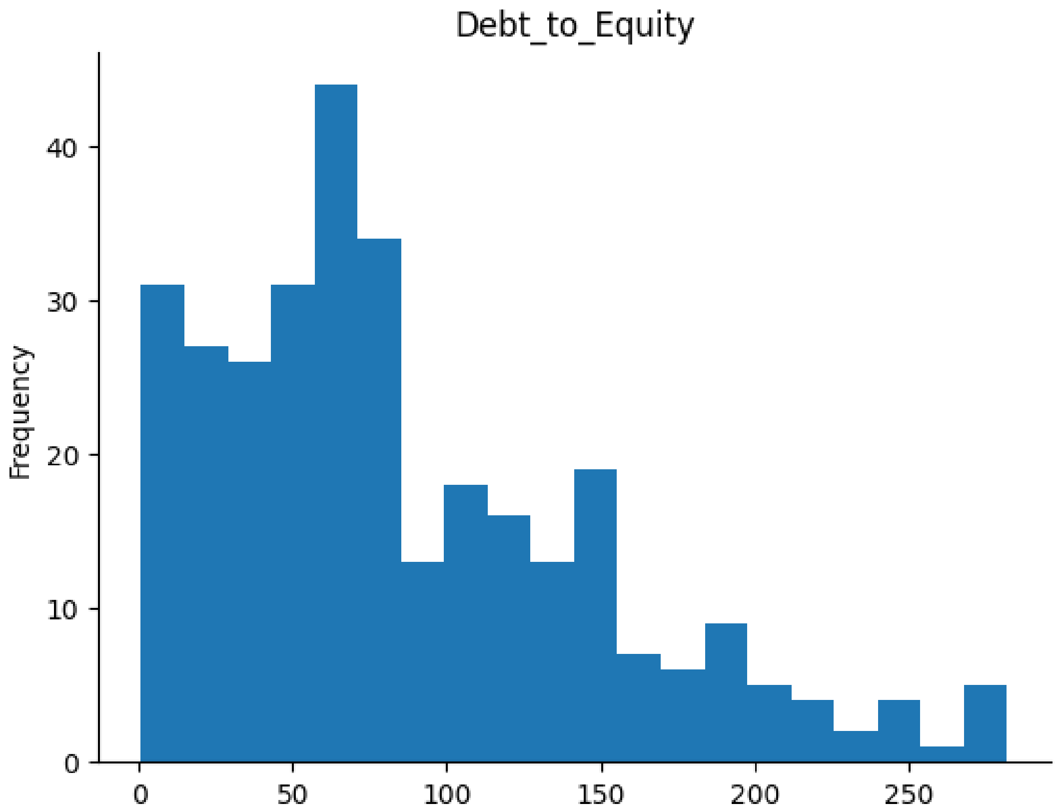
# 5. EXPORT CSV (Session 1: to_csv)

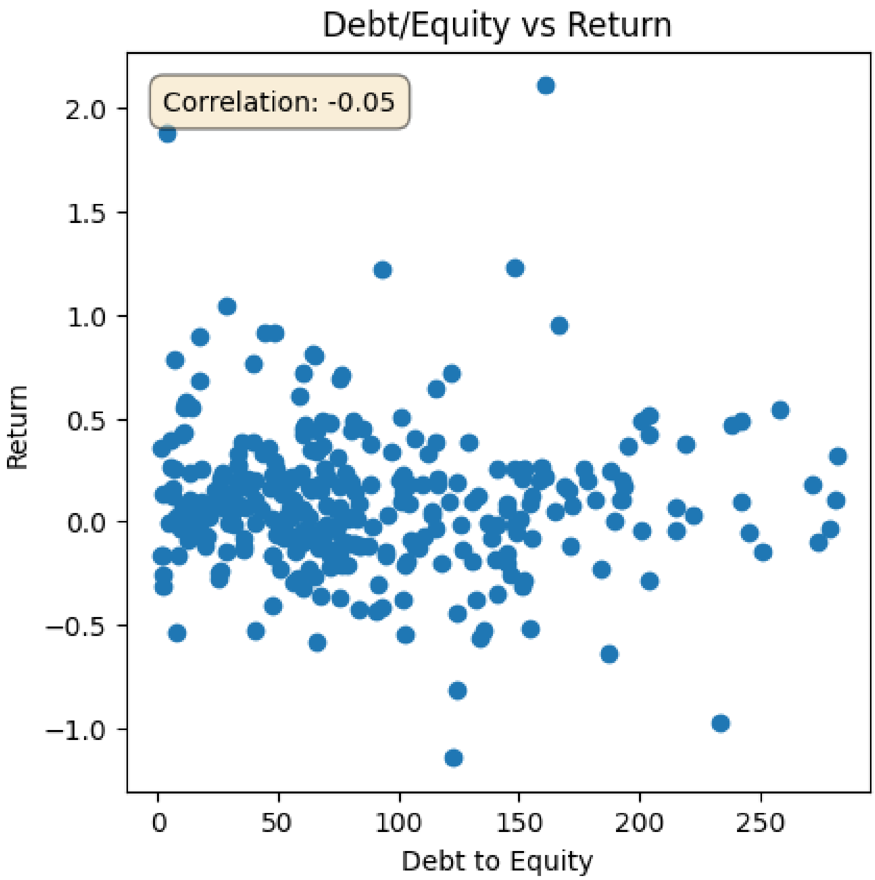
df_clean.to_csv("DHS4_output_cleaned.csv", index=False)
print("  CSV exporté: DHS4_output_cleaned.csv")

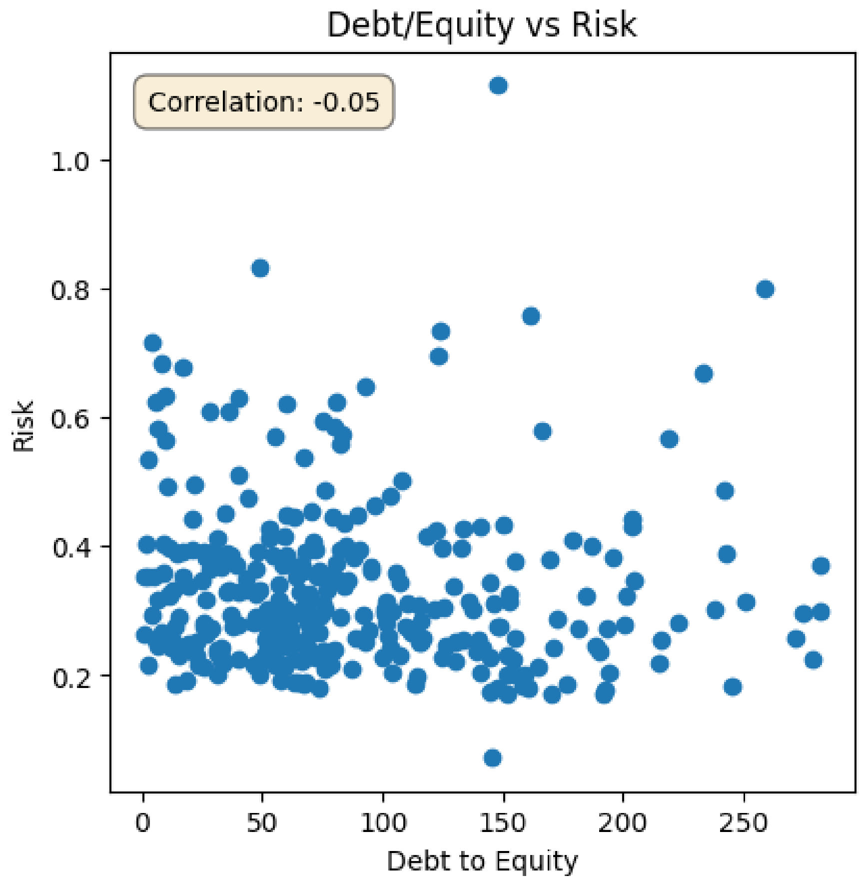
# 6. Final Preview
df_clean
```



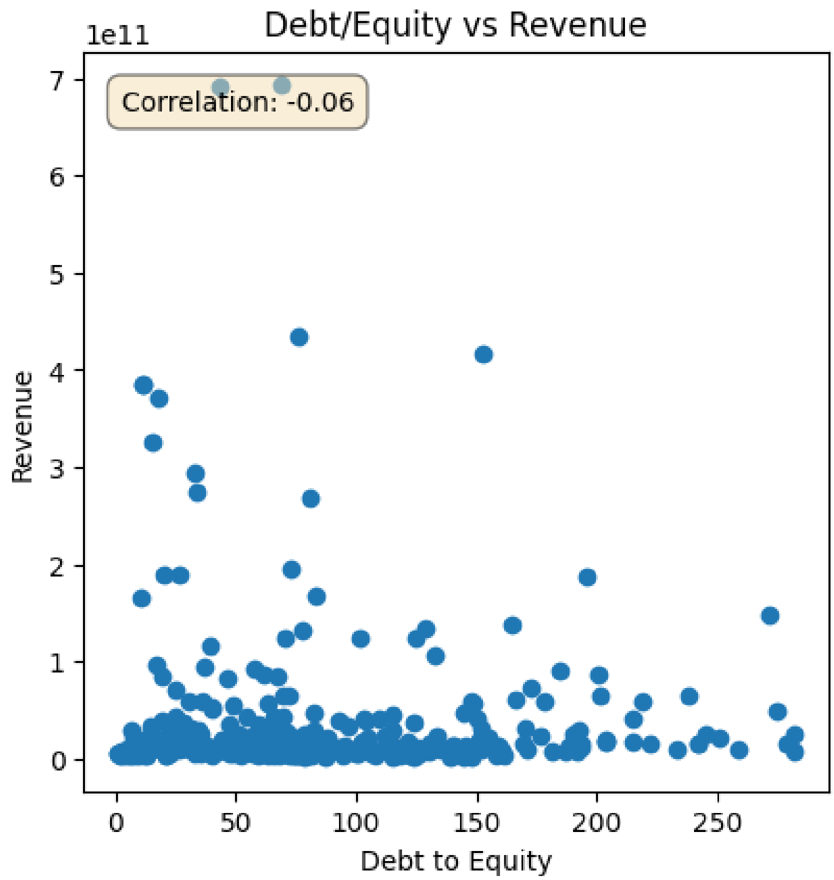


Debt_to_Equity
Frequency
0
10
20
30
40
0
50
100
150
200
250


Debt/Equity vs Return
Correlation: -0.05
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
Return
0
50
100
150
200
250
Debt to Equity



Debt/Equity vs Risk
Correlation: -0.05
Risk
Debt to Equity
1.0
0.8
0.6
0.4
0.2
0
50
100
150
200
250




```
       Debt_to_Equity        Risk      Return       Revenue
count      344.000000  344.000000  344.000000  3.440000e+02
mean       129.164189    0.336358    0.095321  3.928071e+10
std        181.623974    0.134917    0.359176  8.062123e+10
min          0.887000    0.072489   -1.143474  9.405610e+08
25%         45.656000    0.244614   -0.108462  7.490325e+09
50%         75.621000    0.302432    0.063679  1.436450e+10
75%        146.734500    0.388337    0.239145  3.237450e+10
max       1888.690000    1.114902    2.111721  6.931530e+11
 CSV exporté: DHS4_output_cleaned.csv
```

```
[ ]:     Ticker  Debt_to_Equity      Risk    Return       Revenue
     0        A          56.421  0.316108  0.093843  6.788000e+09
     2      AAP         258.405  0.800430  0.540952  8.624327e+09
     3     AAPL         152.411  0.326409  0.251890  4.161610e+11
     6     ABNB          29.324  0.390401 -0.002167  1.158000e+10
     7      ABT          25.310  0.213564  0.093911  4.384300e+10
     ..     …             …         …        …           …
     394    XEL         159.282  0.200991  0.266672  1.422800e+10
```

```
395    XOM          15.672  0.231639  0.020747  3.262450e+11
396    XYL          18.253  0.246674  0.254774  8.894000e+09
398    ZBH          61.603  0.264256 -0.038013  7.833800e+09
399   ZBRA          62.984  0.446898 -0.233670  5.255000e+09

[315 rows x 5 columns]
```

[ ]:
```
from matplotlib import pyplot as plt
df_clean['Debt_to_Equity'].plot(kind='hist', bins=20, title='Debt_to_Equity')
plt.gca().spines[['top', 'right',]].set_visible(False)
```